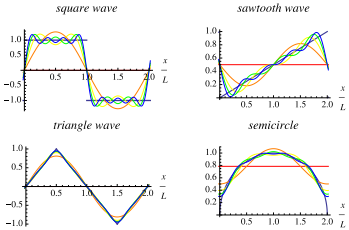
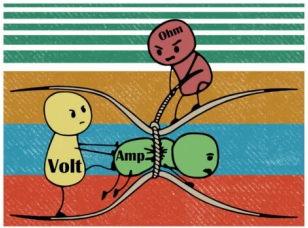
Wk14L1: Fourier

Giles Blaney, PhD





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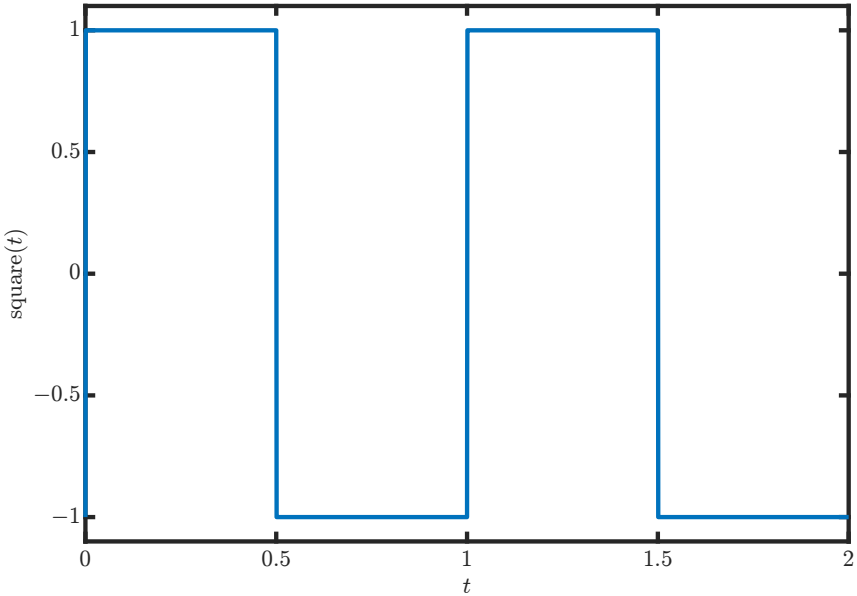
Recap Quiz IV

Fourier's theorem

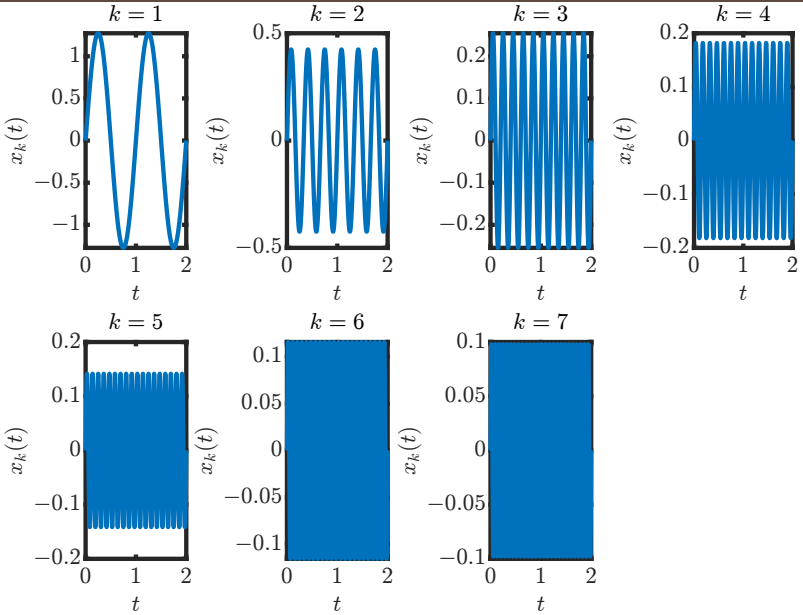
A periodic function can be represented by a sum of sinusoidal harmonics

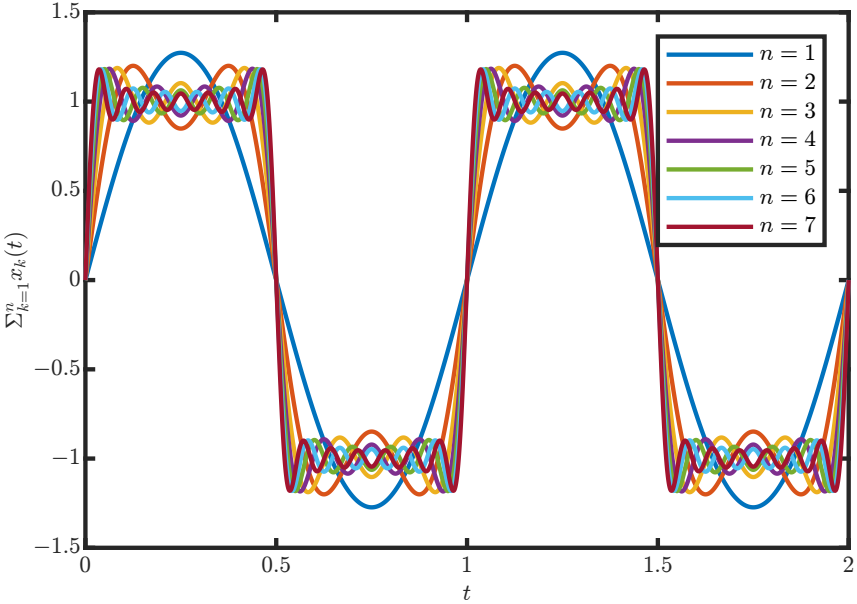
$$f(t) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right)$$


1
0.5
0
−0.5
−1
square(t)
0
0.5
1
1.5
2
t

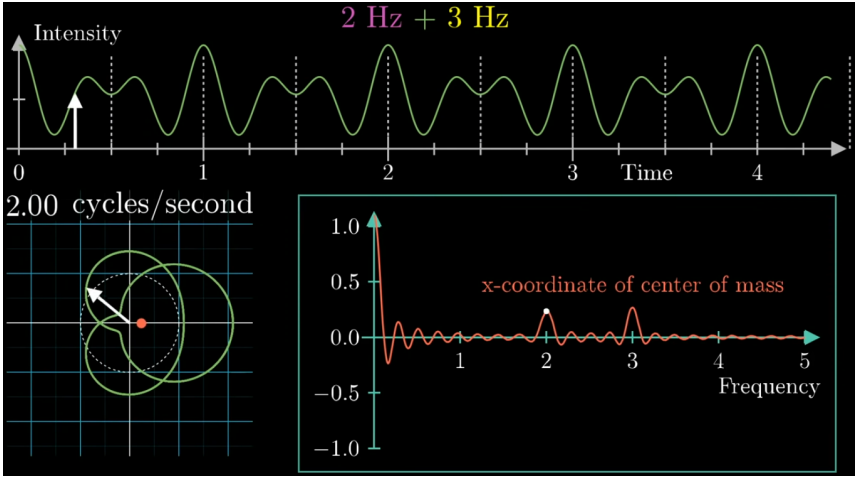
$$\text{square}(t) = \sum_{k=1}^{\infty} x_k(t)$$

$$x_k(t) = \frac{4\sin\left(2\pi\left(2k-1\right)t\right)}{\pi\left(2k-1\right)}$$


$k = 1$
$k = 2$
$k = 3$
$k = 4$
$k = 5$
$k = 6$
$k = 7$
$x_k(t)$
t


1.5
1
0.5
0
−0.5
−1
−1.5
0
0.5
1
1.5
2
t
$\Sigma_{k=1}^{n} x_k(t)$
$n = 1$
$n = 2$
$n = 3$
$n = 4$
$n = 5$
$n = 6$
$n = 7$

Video



youtu.be/spUNpyF58BY?si=gPPAB9EfYB-t3Q92

Fourier Transform

$$\widetilde{F}(\omega) = \mathfrak{F}[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

$$f(t) = \mathfrak{F}^{-1}[\widetilde{F}(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widetilde{F}(\omega)e^{j\omega t}d\omega$$

Table 13-4: Examples of Fourier transform pairs. Note that constant $a \geq 0$.

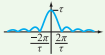
	$f(t)$		$\mathbf{F}(\omega) = \mathcal{F}[f(t)]$	$\lvert\mathbf{F}(\omega)\rvert$
	BASIC FUNCTIONS			
1.		$\delta(t)$	$\longleftrightarrow$ 1	
1a.		$\delta(t - t_0)$	$\longleftrightarrow$ $e^{-j\omega t_0}$	
2.		1	$\longleftrightarrow$ $2\pi\ \delta(\omega)$	
3.		$u(t)$	$\longleftrightarrow$ $\pi\ \delta(\omega) + 1/j\omega$	
4.		$\mathrm{sgn}(t)$	$\longleftrightarrow$ $2/j\omega$	
5.		$\mathrm{rect}(t/\tau)$	$\longleftrightarrow$ $\tau\ \mathrm{sinc}(\omega\tau/2)$	
6.		$\lvert t\rvert$	$\longleftrightarrow$ $-2/\omega^2$	
7.		$e^{-at}\ u(t)$	$\longleftrightarrow$ $1/(a + j\omega)$	
8.		$\cos\omega_0 t$	$\longleftrightarrow$ $\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	
9.		$\sin\omega_0 t$	$\longleftrightarrow$ $j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$	
	ADDITIONAL FUNCTIONS			
10.		$e^{j\omega_0 t}$	$\longleftrightarrow$ $2\pi\ \delta(\omega - \omega_0)$	
11.		$te^{-at}\ u(t)$	$\longleftrightarrow$ $1/(a + j\omega)^2$	
12.		$[e^{-at}\sin\omega_0 t]\ u(t)$	$\longleftrightarrow$ $\omega_0/[(a + j\omega)^2 + \omega_0^2]$	
13.		$[e^{-at}\cos\omega_0 t]\ u(t)$	$\longleftrightarrow$ $(a + j\omega)/[(a + j\omega)^2 + \omega_0^2]$	

Table 13-5: Major properties of the Fourier transform.

Property	$f(t)$		$\mathbf{F}(\omega) = \mathcal{F}[f(t)]$
1. Multiplication by a constant	$K\ f(t)$	$\longleftrightarrow$	$K\ \mathbf{F}(\omega)$
2. Linearity	$K_1\ f_1(t) + K_2\ f_2(t)$	$\longleftrightarrow$	$K_1\ \mathbf{F}_1(\omega) + K_2\ \mathbf{F}_2(\omega)$
3. Time scaling	$f(at)$	$\longleftrightarrow$	$\frac{1}{\lvert a\rvert}\mathbf{F}\left(\frac{\omega}{a}\right)$
4. Time shift	$f(t - t_0)$	$\longleftrightarrow$	$e^{-j\omega t_0}\ \mathbf{F}(\omega)$
5. Frequency shift	$e^{j\omega_0 t}\ f(t)$	$\longleftrightarrow$	$\mathbf{F}(\omega - \omega_0)$
6. Time 1st derivative	$f' = \frac{df}{dt}$	$\longleftrightarrow$	$j\omega\ \mathbf{F}(\omega)$
7. Time nth derivative	$\frac{d^n f}{dt^n}$	$\longleftrightarrow$	$(j\omega)^n\ \mathbf{F}(\omega)$
8. Time integral	$\int_{-\infty}^{t} f(t)\ dt$	$\longleftrightarrow$	$\frac{\mathbf{F}(\omega)}{j\omega} + \pi\ \mathbf{F}(0)\ \delta(\omega)$
9. Frequency derivative	$t^n\ f(t)$	$\longleftrightarrow$	$(j)^n \frac{d^n\mathbf{F}(\omega)}{d\omega^n}$
10. Modulation	$\cos\omega_0 t\ f(t)$	$\longleftrightarrow$	$\frac{1}{2}[\mathbf{F}(\omega - \omega_0) + \mathbf{F}(\omega + \omega_0)]$
11. Convolution in t	$f_1(t) * f_2(t)$	$\longleftrightarrow$	$\mathbf{F}_1(\omega)\ \mathbf{F}_2(\omega)$
12. Convolution in ω	$f_1(t)\ f_2(t)$	$\longleftrightarrow$	$\frac{1}{2\pi}\mathbf{F}_1(\omega) * \mathbf{F}_2(\omega)$

Reminders

- HW 10 Posted
- Quiz V Wednesday